Lecture 3: Limit of Sequence

Mathematical Analysis

1 Convergent Sequences

Definition 1.1 (Convergent Sequence)**.** [def:convergent-sequence] A sequence $a : \mathbb{N}_1 \to F$, where F is a ordered field, is said to be *convergent* if

$$\exists \ell \in F,\, \forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) < \varepsilon.$$

Remark 1.1 (The ε-N Challenge). [rem:epsilon-n-challenge] A value ℓ passes the ε-N challenge if for any proposed $\varepsilon > 0$, there exists an index $N \in \mathbb{N}_1$ such that for all $n \in \mathbb{N}_1$ with $n \geq N$, $d(a_n, \ell) < \varepsilon$.

Remark 1.2. In the The ε-N Challenge, the value of N may depend on the choice of ε. And the value N may be not unique (ie, $N' > N$ also works)

Theorem 1.1 (Uniqueness of Limit)**.** [thm:limit-uniqueness] *Let F be an ordered field, $a : \mathbb{N}_1 \to F$, and $\ell_1, \ell_2 \in F$. Then*

$$\begin{gathered}\big(\forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell_1) < \varepsilon\big) \\ \wedge \\ \big(\forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell_2) < \varepsilon\big) \\ \implies \ell_1 = \ell_2.\end{gathered}$$

Proof. Assume for contradiction that $\ell_1 \neq \ell_2$. Set

$$\varepsilon = \frac{d(\ell_1, \ell_2)}{2}.$$

Since $\ell_1 \neq \ell_2$, we have $\varepsilon > 0$.
By the hypothesis for ℓ_1, there exists $N_1 \in \mathbb{N}_1$ such that for all $n \geq N_1$, $d(a_n, \ell_1) < \varepsilon$. Similarly, there exists $N_2 \in \mathbb{N}_1$ such that for all $n \geq N_2$, $d(a_n, \ell_2) < \varepsilon$.
Let $N = \max(N_1, N_2)$. For any $n \geq N$, both hypotheses hold:

$$d(a_n, \ell_1) < \varepsilon \quad \text{and} \quad d(a_n, \ell_2) < \varepsilon.$$

By the triangle inequality for distance applied via the intermediate point a_n,

$$d(\ell_1, \ell_2) \leq d(\ell_1, a_n) + d(a_n, \ell_2) = d(a_n, \ell_1) + d(a_n, \ell_2),$$

where the last equality uses symmetry of distance. Combining,

$$d(\ell_1, \ell_2) \leq d(a_n, \ell_1) + d(a_n, \ell_2) < 2\varepsilon = d(\ell_1, \ell_2),$$

a contradiction. Hence $\ell_1 = \ell_2$. □

Remark 1.3 (Compact Form Using $\exists!$). [rem:limit-uniqueness-compact] Combined with the existence from the definition of convergent sequence, this theorem states: if a is convergent, then there exists a unique such ℓ:

$$\exists! \ell \in F,\, \forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) < \varepsilon.$$

Here $\exists!$ is shorthand for "existence and uniqueness"; it should be formally introduced in `lec01` as a FOL convention before being used.

2 Convergence Challenge Functions

Definition 2.1 (Convergence Challenge Function)**.** [def:convergence-challenge-function] Let F be an ordered field. A function $g : F \to F$ is a *convergence challenge function* if it satisfies two conditions:

- *Positivity on positives*: $\forall \gamma \in F,\ \gamma > 0 \implies g(\gamma) > 0$.
- *Arbitrarily small values*: $\forall \delta \in F,\ \delta > 0,\ \exists \gamma \in F,\ \gamma > 0 \wedge g(\gamma) < \delta$.

Remark 2.1 (Intuition: Covering Arbitrarily Small Positives)*.* [rem:challenge-intuition]

Theorem 2.1 (Convergence Challenge Replacement)**.** [thm:convergence-challenge-replacement] *Let F be an ordered field, $a : \mathbb{N}_1 \to F$, and let g be a convergence challenge function. Then*

$$\big(\exists \ell \in F,\ \forall \varepsilon \in F,\ \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ \forall n \in \mathbb{N}_1,\ n \geq N,\ d(a_n, \ell) < \varepsilon\big)$$

$$\iff$$

$$\big(\exists \ell \in F,\ \forall \varepsilon \in F,\ \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ \forall n \in \mathbb{N}_1,\ n \geq N,\ d(a_n, \ell) < g(\varepsilon)\big).$$

Proof. Fix the witness $\ell \in F$ in either direction.
($\Rightarrow$) For $\varepsilon > 0$, the positivity-on-positives condition gives $g(\varepsilon) > 0$. Apply the definition of convergent sequence at the challenge $g(\varepsilon)$.
($\Leftarrow$) For $\varepsilon > 0$, the challenge-function property of g gives $\gamma > 0$ with $g(\gamma) < \varepsilon$. The hypothesis at γ gives N such that

$$d(a_n, \ell) < g(\gamma) < \varepsilon \quad \text{for all } n \geq N.$$

□

Lemma 2.1 (Linear Scaling)**.** [lem:linear-is-challenge] *For any $c \in F$ with $c > 0$, the function $g(\gamma) = c\gamma$ is a convergence challenge function.*

Proof. We verify both conditions.
Positivity on positives. For any $\gamma \in F$ with $\gamma > 0$, both $c > 0$ and $\gamma > 0$, so $g(\gamma) = c\gamma > 0$.
Arbitrarily small values. Fix any $\delta \in F$ with $\delta > 0$. We seek $\gamma \in F$ with $\gamma > 0$ and $g(\gamma) < \delta$.
Take $\gamma := \delta/(c+1)$, which is derived in remark 2.2. Since $\delta > 0$ and $c + 1 > 0$ (because $c > 0$), we have $\gamma > 0$.
Now $g(\gamma) = c\gamma = c\delta/(c+1)$. Since $c < c + 1$, we have $c/(c+1) < 1$, hence $c\delta/(c+1) < \delta$. □

Remark 2.2 (Deriving the Witness for Linear Scaling)*.* [rem:linear-scaling-witness-derivation] To find the witness γ used in the proof of lemma 2.1, we start from the target inequality $g(\gamma) < \delta$ and manipulate it to isolate γ:

$$g(\gamma) < \delta \iff c\gamma < \delta \iff \gamma < \frac{\delta}{c}.$$

Any γ in the open interval $\left(0, \frac{\delta}{c}\right)$ will therefore satisfy $g(\gamma) < \delta$. To pick a concrete such γ, we use the fact that $c + 1 > c$, which gives

$$\frac{\delta}{c+1} < \frac{\delta}{c}.$$

Hence we may take $\gamma := \delta/(c+1)$ as the witness.

Remark 2.3 (Why $c \leq 0$ is Excluded)*.* [rem:linear-scaling-cnonpos] If $c \leq 0$, then for any $\gamma \in F$ with $\gamma > 0$, $g(\gamma) = c\gamma \leq 0$. This violates the positivity-on-positives condition, so g is not a convergence challenge function.

Lemma 2.2 (Power). [lem:power-is-challenge] *For any positive integer k, the function $g(\gamma) = \gamma^k$ is a convergence challenge function.*

Proof. Both conditions follow by induction on k. Positivity: $\gamma^k > 0$ for $\gamma > 0$. For arbitrarily small values, take $\gamma := \min(\delta/2, 1/2)$, so $\gamma > 0$ and $\gamma \leq 1$; induction gives $\gamma^k \leq \gamma \leq \delta/2 < \delta$. □

Lemma 2.3 (Summation). [lem:sum-is-challenge] *Let f, g be convergence challenge functions, and suppose both are monotone increasing (i.e. $x \leq y \implies f(x) \leq f(y)$, and similarly for g). Then $f + g$, defined by $(f + g)(\gamma) := f(\gamma) + g(\gamma)$, is a convergence challenge function.*

Proof. Positivity is immediate from $f(\gamma), g(\gamma) > 0$. For $\delta > 0$, pick $\gamma_1, \gamma_2 > 0$ with $f(\gamma_1), g(\gamma_2) < \delta/2$; set $\gamma := \min(\gamma_1, \gamma_2)$. Monotonicity gives $(f + g)(\gamma) \leq f(\gamma_1) + g(\gamma_2) < \delta$. □

Lemma 2.4 (Composition). [lem:challenge-composition] *Let f, g be convergence challenge functions, and suppose f is monotone increasing (i.e. $x \leq y \implies f(x) \leq f(y)$). Then $f \circ g$ is a convergence challenge function.*

Proof. Positivity: $f(g(\gamma)) > 0$ since $g(\gamma) > 0$ and f preserves positivity. For $\delta > 0$, pick $\eta > 0$ with $f(\eta) < \delta$, then $\gamma > 0$ with $g(\gamma) < \eta$. Monotonicity of f gives $(f \circ g)(\gamma) \leq f(\eta) < \delta$. □

Theorem 2.2 (Closed Form of Convergence). [thm:convergence-closed-form] *Let F be an ordered field, let $a : \mathbb{N}_1 \to F$. Then*

$$\big(\exists \ell \in F,\, \forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) < \varepsilon\big)$$

$$\iff$$

$$\big(\exists \ell \in F,\, \forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) \leq \varepsilon\big).$$

Proof. ($\Rightarrow$) If $d(a_n, \ell) < \varepsilon$, then $d(a_n, \ell) \leq \varepsilon$.
($\Leftarrow$) By lemma 2.1 with $c = 2$, $g(\varepsilon) = 2\varepsilon$ is a convergence challenge function. From the closed-form hypothesis, for every $\varepsilon > 0$ there exists N with $d(a_n, \ell) \leq \varepsilon$ for all $n \geq N$. Since $\varepsilon > 0 \implies \varepsilon < 2\varepsilon$, we obtain $d(a_n, \ell) \leq \varepsilon < 2\varepsilon$, hence

$$\exists \ell \in F,\, \forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) < 2\varepsilon. \tag{$*$}$$

For any $\varepsilon > 0$, applying $(*)$ at $\varepsilon/2 > 0$ gives N such that $d(a_n, \ell) < 2 \cdot (\varepsilon/2) = \varepsilon$ for all $n \geq N$. □

3 Limit of a Sequence

Definition 3.1 (Limit of a Sequence). [def:limit] Let F be an ordered field. We say the sequence $a : \mathbb{N}_1 \to F$ *has a limit of* $\ell \in F$ if

$$\forall \varepsilon \in F,\, \varepsilon > 0,\, \exists N \in \mathbb{N}_1,\, \forall n \in \mathbb{N}_1,\, n \geq N,\, d(a_n, \ell) < \varepsilon.$$

Redirect (Converges To) [def:converges-to]
Moved to definition 3.1.

Definition 3.3 (Set of Convergent Sequences). [def:convergent-sequences-set] Let F be an ordered field. The *set of convergent sequences in F* is

$$\mathrm{Conv}(F) := \{a : \mathbb{N}_1 \to F \mid a \text{ is convergent}\}.$$

Definition 3.4 (Limit Operator). [def:limit-op] Let F be an ordered field. The *limit operator* is the function

$$\lim : \mathrm{Conv}(F) \to F, \quad a \mapsto \text{the unique } \ell \in F \text{ such that } a\text{has a limit of}\ell.$$

We write

$$\lim a_n = \ell.$$

Redirect (Limit of a Sequence) [def:limit-of-sequence]
Moved to definition 3.4.

Remark 3.1 (Well-Definedness of the Limit). [rem:limit-well-defined] The function lim in definition 3.4 is well-defined for two reasons:

- *Every input has an output*: for every $a \in \mathrm{Conv}(F)$, the definition of convergent sequence guarantees some $\ell \in F$ with $a$has a limit ofℓ.
- *Every input has exactly one output*: by theorem 1.1, this ℓ is unique.

Theorem 3.1 (Scalar Multiplication of the Limit). [thm:limit-scalar-multiplication] *Let F be an ordered field, let $a \in \mathrm{Conv}(F)$, and let $c \in F$. Then*

$$\lim(c \cdot a_n) = c \cdot \lim a_n.$$

Theorem 3.2 (Addition of the Limit). [thm:limit-addition] *Let F be an ordered field, and let $a, b \in \mathrm{Conv}(F)$. Then*

$$\lim(a_n + b_n) = \lim a_n + \lim b_n.$$

Theorem 3.3 (Linearity of the Limit). [thm:limit-linearity] *Let F be an ordered field, and let $a, b \in \mathrm{Conv}(F)$. Then*

$$\lim(c \cdot a_n + d \cdot b_n) = c \cdot \lim a_n + d \cdot \lim b_n.$$

Theorem 3.4 (Multiplication of the Limit). [thm:limit-multiplication] *Let F be an ordered field, and let $a, b \in \mathrm{Conv}(F)$. Then*

$$\lim(a_n \cdot b_n) = \lim a_n \cdot \lim b_n.$$

Theorem 3.5 (Division of the Limit). [thm:limit-division] *Let F be an ordered field, and let $a, b \in \mathrm{Conv}(F)$ with $\forall n \in \mathbb{N}_1, b_n \neq 0$ and $\lim b_n \neq 0$. Then*

$$\lim\left(\frac{a_n}{b_n}\right) = \frac{\lim a_n}{\lim b_n}.$$

Theorem 3.6 (Order Preservation of Limits). [thm:limit-order-preservation] *Let F be an ordered field, and let $a, b \in \mathrm{Conv}(F)$ with $\forall n \in \mathbb{N}_1$, $a_n \leq b_n$. Then*

$$\lim a_n \leq \lim b_n.$$

4 Bounded Sequences

Definition 4.1 (Bounded Sequence). [def:bounded-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *bounded* if

$$\exists M \in F,\, \forall n \in \mathbb{N}_1,\, d(a_n, 0) \leq M$$

Theorem 4.1 (Convergent Sequences are Bounded). [thm:convergent-sequences-are-bounded] *If a sequence $a : \mathbb{N}_1 \to F$ is convergent, it must be bounded. ie,*

$$(\exists \ell \in F, \forall \varepsilon \in F > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1 > N, d(a_n, \ell) < \varepsilon) \implies (\exists M \in F, \forall n \in \mathbb{N}_1, d(a_n, 0) \leq M)$$

Proof. Suppose the convergent sequence a, $\lim a = \ell$.
Take $\varepsilon = 1$, then there exists $N \in \mathbb{N}_1$ such that for all $n \in \mathbb{N}_1$ with $n > N$, $d(a_n, \ell) < 1$.
According to the triangle inequality for distance, and $d(a_n, \ell) < 1$, we have

$$d(0, a_n) \leq d(0, \ell) + d(\ell, a_n) < d(0, \ell) + 1$$

Which means for all $n \in \mathbb{N}_1$ with $n > N$, $d(0, a_n) < d(0, \ell) + 1$.
And for $n \in \mathbb{N}_1$ with $n \leq N$, we have $d(0, a_n) \leq \max\{d(0, a_1), d(0, a_2), ..., d(0, a_N)\}$.
So, we can take $M = \max\{\max\{d(0, a_1), d(0, a_2), ..., d(0, a_N)\}, d(0, \ell) + 1\}$. Hence M exists for all convergent sequences a. □

Remark 4.1. Although Convergent $\implies$ Bounded, the converse is not true. A bounded sequence may not be convergent.
For example, the sequence $a_n = (-1)^n$ is bounded (since $d(a_n, 0) = 1$ for all n), but not convergent.

5 Monotone Convergence Theorem

Definition 5.1 (Increasing Sequence)**.** [def:increasing-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *increasing* if

$$\forall n \in \mathbb{N}_1, a_n \leq a_{n+1}$$

Definition 5.2 (Strictly Increasing Sequence)**.** [def:strictly-increasing-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *strictly increasing* if

$$\forall n \in \mathbb{N}_1, a_n < a_{n+1}$$

Definition 5.3 (Decreasing Sequence)**.** [def:decreasing-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *decreasing* if

$$\forall n \in \mathbb{N}_1, a_n \geq a_{n+1}$$

Definition 5.4 (Strictly Decreasing Sequence)**.** [def:strictly-decreasing-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *strictly decreasing* if

$$\forall n \in \mathbb{N}_1, a_n > a_{n+1}$$

Lemma 5.1 (Strictly Increasing Extends to All Indices)**.** [lem:strictly-increasing-extends] *Let $a : \mathbb{N}_1 \to F$ be strictly increasing. Then $\forall n, m \in \mathbb{N}_1$, $n > m \implies a_n > a_m$.*

Lemma 5.2 (Increasing Extends to All Indices)**.** [lem:increasing-extends] *Let $a : \mathbb{N}_1 \to F$ be increasing. Then $\forall n, m \in \mathbb{N}_1$, $n \geq m \implies a_n \geq a_m$.*

Lemma 5.3 (Strictly Decreasing Extends to All Indices)**.** [lem:strictly-decreasing-extends] *Let $a : \mathbb{N}_1 \to F$ be strictly decreasing. Then $\forall n, m \in \mathbb{N}_1$, $n > m \implies a_n < a_m$.*

Lemma 5.4 (Decreasing Extends to All Indices)**.** [lem:decreasing-extends] *Let $a : \mathbb{N}_1 \to F$ be decreasing. Then $\forall n, m \in \mathbb{N}_1$, $n \geq m \implies a_n \leq a_m$.*

Definition 5.5 (Monotone Sequence)**.** [def:monotone-sequence] A sequence $a : \mathbb{N}_1 \to F$ is said to be *monotone* if it is either increasing or decreasing.

Theorem 5.1 (Monotone Convergence Theorem)**.** [thm:mct] *Let F be an complete ordered field, and let $a : \mathbb{N}_1 \to F$ be a monotone sequence. If a is bounded, then a is convergent.*

Proof. Suppose a is a bounded sequence in F and increasing sequence, which means exist $x = \sup a$ (by the Axiom of Completeness, applied to the nonempty set $\{a_n : n \in \mathbb{N}_1\}$ bounded above in F), and $\forall n, m \in \mathbb{N}_1,\ n \le m \implies a_n \le a_m$.
According to the ε criterion for supremum, $\forall n \in \mathbb{N}_1,\ a_n \le x$ and $\forall \varepsilon \in F,\ \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ a_N > x - \varepsilon$
Adding the increasing property, ie

$$N \le n \implies a_N \le a_n \le x \implies (a_N > x - \varepsilon \implies a_n \ge a_N > x - \varepsilon)$$

We can have

$$\forall \varepsilon \in F,\ \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ \forall n \in \mathbb{N}_1 \ge N,\ a_n > x - \varepsilon$$

Also,

$$a_n \le x \land \varepsilon > 0 \implies a_n < x + \varepsilon$$

$$a_n > x - \varepsilon \land a_n < x + \varepsilon \implies x - \varepsilon < a_n < x + \varepsilon \implies d(a_n, x) < \varepsilon$$

Notice that such x is the supremum of the upper bounded sequence a, it exists.

$$\exists x \in F,\ \forall \varepsilon \in F,\ \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ \forall n \in \mathbb{N}_1 \ge N,\ d(a_n, x) < \varepsilon$$

Which means it is a convergent sequence and its limit is x.
If instead a is decreasing, the same argument applies with $x := \inf a$ (which exists by the Axiom of Completeness, since $\{a_n : n \in \mathbb{N}_1\}$ is nonempty and bounded below), using the dual ε-criterion for the infimum. □

6 Comparison of Sequences

Theorem 6.1 (Comparison for Monotone Sequences)**.** [thm:monotone-comparison-increasing] *Let F be a complete ordered field, and let $(s_n), (t_n) : \mathbb{N}_1 \to F$ be increasing with $s_n \le t_n$ for all $n \in \mathbb{N}_1$. If (t_n) is convergent, then*

$$(s_n) \text{ convergent and } \lim s_n \le \lim t_n.$$

Proof. **TODO** □

Theorem 6.2 (Comparison for Decreasing Sequences)**.** [thm:monotone-comparison-decreasing] *Let F be a complete ordered field, and let $(s_n), (t_n) : \mathbb{N}_1 \to F$ be decreasing with $s_n \le t_n$ for all $n \in \mathbb{N}_1$. If (s_n) is convergent, then*

$$(t_n) \text{ convergent and } \lim s_n \le \lim t_n.$$

Proof. Apply theorem 6.1 to the increasing sequences $(-t_n), (-s_n)$. □

7 Squeeze Theorem

Theorem 7.1 (Squeeze Theorem)**.** [thm:squeeze-theorem] *Let F be an ordered field, and let $a, x, b : \mathbb{N}_1 \to F$ with $a_n \le x_n \le b_n$ for all $n \in \mathbb{N}_1$. If a and b are convergent with $\lim a_n = \lim b_n = L$, then x is convergent and $\lim x_n = L$.*

Proof. **TODO** □